FIITJEE

ALL INDIA TEST SERIES

FULL TEST – XVI

JEE (Main)-2022

TEST DATE: 07-07-2022

Time Allotted: 3 Hours **Maximum Marks: 300**

General Instructions:

- The test consists of total 90 questions.
- Each subject (PCM) has 30 questions.
- This question paper contains **Three Parts.**
- **Part-A** is Physics, **Part-B** is Chemistry and **Part-C** is Mathematics.
- Each part has only two sections: **Section-A and Section-B**.
- **Section – A** : Attempt all questions.
- **Section – B :** Do any five questions out of 10 Questions.

Section-A (01 – 20, 31 – 50, 61 – 80) *contains 60 multiple choice questions which have **only one correct answer**. Each question carries **+4 marks** for correct answer and **–1 mark** for wrong answer.*

***Section-B (21 – 30, 51 – 60, 81 – 90)** contains 30 Numerical answer type questions with answer XXXXX.XX and each question carries **+4 marks** for correct answer and **–1 mark** for wrong answer.*

Physics

PART – A

SECTION – A
(One Options Correct Type)

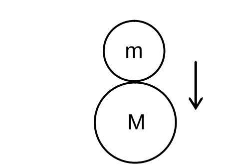
This section contains **20 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

1. A person walking at the rate of 3km/hour, the rain appears to fall vertically when he increase his to speed 6 km/hr it appears to meet him at angle of 45^0 with vertical. The speed of rain is
   (A) $3\sqrt{2}\ \text{km/hr}$
   (B) $\frac{3}{\sqrt{2}}\ \text{km/hr}$
   (C) $6\sqrt{2}\ \text{km/hr}$
   (D) $2\sqrt{3}\ \text{km/hr}$

2. Force acting on a particle moving in a straight line varies with the velocity of the particle as $F = \frac{K}{v}$. Here, K is a constant. The work done by the this force in time t is:
   (A) $\frac{K}{v^2}.t$
   (B) 2Kt
   (C) Kt
   (D) $\frac{2Kt}{v^2}$

3. A spring has a length l_1 when tension in it is n_1(in N). It has a length l_2 when tension is n_2 (in N). Find its spring constant :
   (A) $\frac{(n_2l_2 - n_1l_1)}{(l_1 - l_2)}$
   (B) $\frac{(n_1 - n_2)}{(l_1 - l_2)}$
   (C) $\frac{(n_2 - n_1)}{(l_1 - l_2)}$
   (D) $\frac{(n_1l_1 - n_2l_2)}{(l_1 - l_2)}$

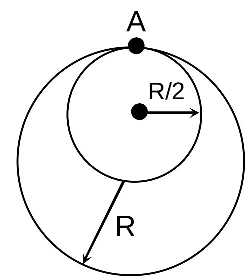
4. A small ball of mass m is placed on top of the "super ball" of mass M and the two ball are dropped to the floor from height h. Height rise of small ball after the collision is Nh, then 'N' is: (Assume that the collision with the superball are elastic, and the m << M)



   (A) 8
   (B) 9
   (C) 10
   (D) 11

5. A spherical object of mass 1 kg and radius 1 m is falling vertically downward inside a viscous liquid in a gravity free space. At a certain instant the velocity of the sphere is 2 m/s. If the coefficient of viscosity of the liquid is $\frac{1}{18\pi}$ N-S/m^2, then velocity of ball will become 0.5 m/s after a time
(A) ln 4 s
(B) 2 ln 4 s
(C) 3 ln 4 s
(D) 2 ln 2 s

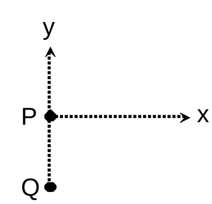
6. Two rings made of same material and thickness, one of radius R and other of radius R/2 are welded together at point A. Now, it is hanged on a nail at wall, the nail touching both the rings at A. Now it is slightly displaced in the plane of rings and released. The period of small oscillations is



(A) $2\pi\sqrt{\frac{2R}{5g}}$
(B) $2\pi\sqrt{\frac{5R}{6g}}$
(C) $2\pi\sqrt{\frac{9R}{5g}}$
(D) $2\pi\sqrt{\frac{5R}{2g}}$

7. The efficiency of carnot engine is 0.6. It rejects 20 J of heat to the sink. The work done by the engine is
(A) 20 J
(B) 30 J
(C) 33.5 J
(D) 50 J

8. Two identical sources P and Q emit waves in same phase and of same wavelength. Spacing between P and Q is 3λ. The maximum distance from P along the x–axis at which a minimum intensity occurs is given by



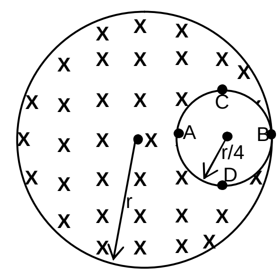
(A) 6.58λ
(B) 2.25λ
(C) 8.75λ
(D) 0.55λ

9. A boat which has a speed of 6 km/h in still water crosses a river of width 1 km along the shortest possible path in 20 min. The velocity of the river water in km/h is
(A) 1
(B) 3
(C) 4
(D) $3\sqrt{3}$

10. If photons of energy 12.75 eV are passing through hydrogen gas in ground state then total possible number of lines in emission spectrum will be
(A) 6
(B) 4
(C) 3
(D) 2

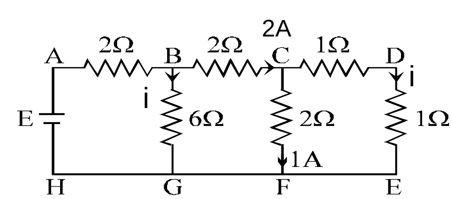
11. The momentum of α-particles moving in a circular path of radius 10 cm in a perpendicular magnetic field of 0.05 Tesla will be:
(A) 1.6×10^{-20} kg m/s
(B) 1.6×10^{-21} kg m/s
(C) 1.6×10^{-19} kg m/s
(D) 1.6×10^{-18} kg m/s

12. Consider a spherical charge distribution of volume charge density ρ c /m^3 Potential difference between A & B



(A) $\frac{2\rho}{3\varepsilon_0} r^2$
(B) 0
(C) $\frac{\rho}{4\varepsilon_0} r^2$
(D) $\frac{\rho r^2}{8\varepsilon_0}$

13. The figure shows a network of resistors and a battery.
If 1A current flows through the branch CF, then the emf E of the battery is

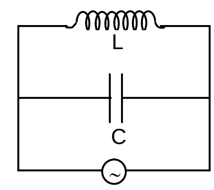


(A) 24 V
(B) 12 V
(C) 18V
(D) 6V

14. The current in a wire varies with time according to the equation I = 4 + 2t, where I is in ampere and t is in sec. The quantity of charge which has passed through a cross-section of the wire during the time t = 2 sec to t = 6 sec will be
(A) 60 coulomb
(B) 24 coulomb
(C) 48 coulomb
(D) 30 coulomb

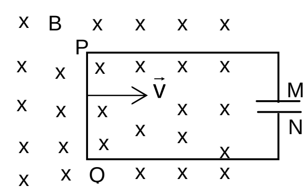
15. If ρ is the density of the material of a wire and σ is the breaking stress. The greatest length of the wire that can hang freely without breaking is
(A) $\frac{2\sigma}{\rho g}$
(B) $\frac{\rho}{\sigma g}$
(C) $\frac{\rho g}{2\sigma}$
(D) $\frac{\sigma}{\rho g}$

16. For the circuit shown in the figure, the current through the inductor is 0.6 A, while the current through the capacitor is 0.4 A. The current drawn from the generator is



(A) 1.0 A
(B) 0.4 A
(C) 0.6 A
(D) 0.2 A

17. A rod PQ is connected to the capacitor plates. The rod is placed in a magnetic field (B) directed downward perpendicular to the plane of the paper. If the rod is pulled out of magnetic field with velocity v as shown in figure.

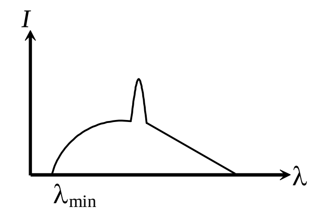


(A) plate M will be positively charged
(B) plate N will be positively charged
(C) both plates will be similarly charged
(D) no charge will be collected on plates

18. A light emitting diode (LED) has a voltage drop of 2V across it and passes a current of 10 mA. When it operates with a 6V battery through a limiting resistor R, the value of R is:
(A) 40 k Ω
(B) 4 k Ω
(C) 200 Ω
(D) 400 Ω

19. The number densities of electrons and holes in a pure germanium at room temperature are equal and its value is 3×10^{16} per m^3. On doping with aluminium, the hole density increases to 4.5×10^{22} per m^3. Then the electron density in doped germanium is
(A) $2.5 \times 10^{10}\ m^{-3}$
(B) $2 \times 10^{10}\ m^{-3}$
(C) $4.5 \times 10^{9}\ m^{-3}$
(D) $3 \times 10^{9}\ m^{-3}$

20. An X-ray tube has three main controls.
(i) the target material (its atomic number Z)
(ii) the filament current (I_f) and
(iii) the accelerating voltage (V)
Figure shows a typical intensity distribution against wavelength. Which of the following is incorrect?



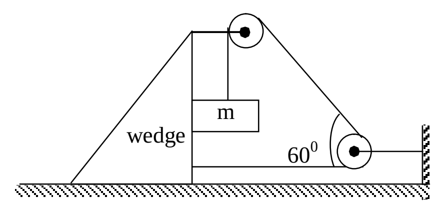
(A) The limit λ_{min} is proportional to V^{-1}
(B) The sharp peak shifts to the right as Z is increased
(C) The penetrating power of X ray increases if V is increased
(D) The intensity everywhere increases if filament current I_f is increased

SECTION – B
(Numerical Answer Type)

This section contains **10** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

21. A thin wire of area of cross-section $A = 10^{-2} m^2$ is used to make a ring of radius $r = 10^{-1} m$. This ring is placed on a smooth horizontal floor & is given angular velocity $\omega = 2$ rad / s about its centre. Find out stress in the ring (mass per unit length of wire λ = 1 kg/m)

22. A wedge and block are connected by a mass less string passing over a frictionless pulley as shown in the figure. At the instant shown, the speed of the wedge is 1m/s. Assume all surfaces are smooth. The speed of the block with respect to ground at the instant shown is



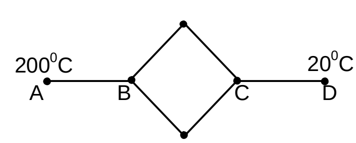
23. A satellite is moving round the earth. In order to escape it the velocity of satellite must be increased by what percent?

24. The displacement x (in m) of particle of mass m (in kg) is related to time t (in second) by $x = t^2 + 3$. Find the work (in J) done in the first two seconds.(take m = 0.5 kg)

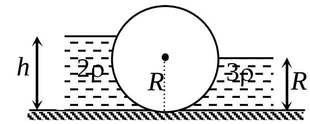
25. A block of mass 1 kg lies on a horizontal surface in a truck. The coefficient of static friction between the block and the surface is 0.6. If the acceleration of the truck is 5 m/s^2, then what frictional force acting on the block (in newton.)

26. A block of silver of mass 4 kg hanging from a string is immersed in a liquid of relative density 0.72. If relative density of silver is 10.5, then tension in the string (in Newton) will be

27. Six identical conducting rods are joined as shown in figure. Points A and D are maintained at constant temperatures of 200^0C and 20^0C respectively. Temperature of junction B in degree celsius will be

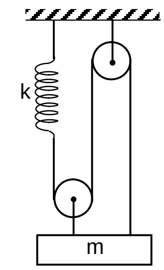


28. In the figure shown, the heavy cylinder (radius R) resting on a smooth surface separates two liquids of densities 2ρ and 3ρ. The height h/R… for the equilibrium of cylinder must be



29. Velocity time equation of a particle moving in a straight line is $V = t^2 - 5t + 6$. The distance travelled by the particle in the time interval from t = 0 to t = 4 sec

30. For a system in equilibrium as shown in figure elongation in spring will be (n mg) /k, then n is



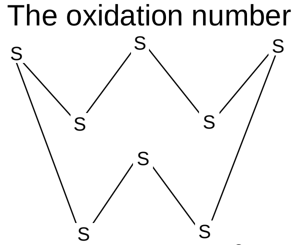
Chemistry

PART – B

SECTION – A
(One Options Correct Type)

This section contains **20 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

31. The oxidation number and hybridization of sulphur in S_8 molecule are respectively:



(A) +2 and sp^2
(B) +2 and sp^3
(C) zero and sp^2
(D) zero and sp^3

32. At high pressure the van der Waal's equation for one mole of a real gas reduces to

(A) $PV = RT - \frac{a}{V}$
(B) PV = RT + Pb
(C) $\left(P + \frac{a}{V^2}\right)V = RT$
(D) PV = RT

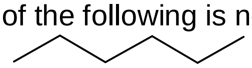
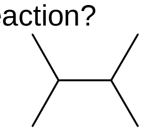
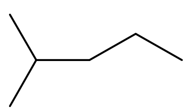
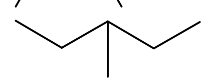
33. Which of the following compound on hydrolysis forms products that are polymerized to silicones?
(A) $SiCl_4$
(B) $(CH_3)_2SiCl_2$
(C) SiH_2Cl_2
(D) H_2SiF_6

34. $X(g) \rightleftharpoons Y(g) + Z(g)$
The equilibrium constant K_p of above reaction is 3 atm at a certain temperature. If equal moles of X, Y and Z are present at equilibrium in a one litre container, the equilibrium pressure would be:
(A) 12 atm
(B) 3 atm
(C) 6 atm
(D) 9 atm

35. Two grams of dihydrogen diffuse from a container in 10 minutes. How many gram of dioxygen would diffuse through the same container in the same time under similar conditions?
(A) 4
(B) 8
(C) 6
(D) 0.5

36. $NaOH + CO \longrightarrow Product$
Which of the following is/are the product(s) of above reaction?
(A) HCOONa
(B) $NaHCO_3$
(C) $Na_2CO_3 + H_2$
(D) $NaH + CO_2$

37. Which of the following aqueous solution is colourless?
(A) $CuSO_4$
(B) $Zn(NO_3)_2$
(C) $FeCl_2$
(D) $KMnO_4$

38. $CH_3CH_2CH_2Cl + CH_3\underset{\underset{Cl}{|}}{C}HCH_3 \xrightarrow[\text{Dry ether}]{\text{Na}} Products$
Which of the following is not a product of above reaction?
(A) 
(B) 
(C) 
(D) 

39. The half-life of a chemical reaction does not change by either increasing or decreasing the initial concentration of reactant. What is the order of the reaction?
(A) Zero
(B) First
(C) Second
(D) Fractional

40. The relation between the molar conductance ($\wedge_m$) and equivalent conductance ($\wedge_e$) of the solution of $MgCl_2.KCl.6H_2O$ is
(A) $\wedge_m = \wedge_e \times 8$
(B) $\wedge_m = \wedge_e \times 3$
(C) $\wedge_e = \wedge_m \times 3$
(D) $\wedge_e = \wedge_m \times 8$

41. The strongest reducing agent out of the following is:
(A) NH_3
(B) PH_3
(C) AsH_3
(D) BiH_3

42. A diene on reductive ozonolysis produces two moles of ethanal and one mole of propan -1, 3-dial. How many geometrical isomer(s) is/are possible for the diene?
(A) 2
(B) 3
(C) 4
(D) 6

43. The hybridization of xenon in XeO_3 is:
(A) sp^3d^2
(B) sp^3d
(C) sp^3d^3
(D) none of these

44. Which of the following has the most acidic hydrogen?
(A) 3-hexanone
(B) 2, 4-hexanedione
(C) 2, 5-hexanedione
(D) 2, 3-hexanedione.

45. Which of the following solutions in 1: 1 molar ratio behave as buffer?
(A) NaOH + KCl
(B) $CH_3COOH + CH_3COONa$
(C) HCl + KCl
(D) $H_2C_2O_4 + Na_2CO_3$

46. The shape of $[Co(NH_3)_6]^{3+}$ is:
(A) Tetrahdedral
(B) Trigonal bipyramidal
(C) Octahedral
(D) Square planar

47. How many effective number of Na^+ ions are present in the unit cell of NaCl if the ions along all the body diagonals of the unit cell are removed?
(A) 4
(B) 3
(C) 2
(D) 1

48. Which one of the following has aromatic character?
(A) Cyclopentadienyl cation
(B) Cyclopentadienyl radical
(C) Cyclopentadienyl anion
(D) Cyclopentadiene

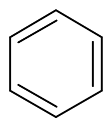
49. Which of the following two reagents form silicates when react with silica (SiO_2)?
(A) NaCl and Na_2S
(B) NaOH and Na_2CO_3
(C) Na_2SO_4 and $NaNO_3$
(D) NaBr and $NaHSO_4$

50. $C_6H_5CHO + CH_3COOCOCH_3 \xrightarrow[\Delta]{CH_3COONa} Products$
If one product of above reaction is $C_6H_5 - CH = CH - COOH$, the other organic product(side product) is:
(A) CH_3CH_2OH
(B) C_6H_5COOH
(C) CH_3CHO
(D) $C_6H_5 - CH = CH_2$

SECTION – B
(Numerical Answer Type)

This section contains **10** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

51. In the following reaction:
$P_4 + xNaOH + yH_2O \longrightarrow PH_3 + 3NaH_2PO_2$
Sum of x + y is

52.  $+ CH_3CH_2CH_2CH_2Cl \xrightarrow{\text{Anhy.}AlCl_3} \text{Products}$

How many monosubstituted product(s) is/are possible for the above reaction?
(Consider stereoisomers)

53. Compound X dissociates according to the reaction $2X(g) \rightleftharpoons 2Y(g) + Z(g)$, with degree of dissociation which is small compared to unity, if the expression for α in terms of equilibrium constant K_P and total pressure P is given as $\alpha = \left(\frac{2K_P}{P}\right)^{1/n}$. The value of n is

54. How many gram of copper will be deposited at cathode by passing 1 Faraday electricity through aqueous $CuSO_4$ solution?
[At. mass of Cu = 63.5]

55. For a reaction $A \rightarrow B$, ΔH = +ve, graph between log K_{eq} and $\frac{1}{T}$ is a straight line of slope $= -\frac{1}{4.606}$. Find ΔH in calories. [R = 2 cal K^{-1} mol^{-1}]

56. What is the effective atomic number of iron in $K_4[Fe(CN)_6]$?

57. Calculate the resonance energy of N_2O in kJ mol^{-1} unit from the following data: (ΔH_f^0 of N_2O = 82 kJ/mol)
$BE_{N \equiv N}$ = 946.2 kJ/mol
$BE_{N = N}$ = 418 kJ/mol
$BE_{O = O}$ = 497 kJ/mol
$BE_{N = O}$ = 605.3 kJ/mol

58. How many maximum number of electrons of an atom have $n + \ell = 4$?
n = Principal quantum number
ℓ = Azimuthal quantum number

59. $B_3N_3H_6 + HCl \longrightarrow B_3N_3H_9Cl_3$
If x = number of N – H bond present in the product.
y = number of N – Cl bonds present in the product
z = number of B – Cl bond present in the product
What is the value of (x + y + z)?

60. A 40% by mass of HNO_3 solution has a density of 1.2 g/mL. If the molarity of the solution is M, then what is the possible value of M.

Mathematics

PART – C

SECTION – A
(One Options Correct Type)

This section contains **20 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

61. If $f(x)=\sqrt{x+3-4\sqrt{x-1}}+\sqrt{x+8-6\sqrt{(x-1)}}$, then $f'(x)$ at $x=1.5$ is
   (A) 0
   (B) $-\sqrt{2}$
   (C) $-\sqrt{3}$
   (D) –4

62. A normal drawn to parabola $y^2 = 4ax$ meet the curve again at Q such that angle subtend by PQ at vertex is 90° then coordinate of P can be
   (A) $(8a, 4\sqrt{2}a)$
   (B) (8a, 4a)
   (C) $(2a, -2\sqrt{2}a)$
   (D) none of these

63. Let $\vec{a}=\hat{i}+\lambda\hat{j}+\hat{k}$ and $\vec{b}=\hat{i}+\hat{j}+\hat{k}$. If $|\vec{a}+\vec{b}|=|\vec{a}|+|\vec{b}|$ and $f(x)=\int_0^{\sqrt{x}}\sin(t^2)\,dt$, then the value of $f'(\lambda)$, is:
   (A) $\frac{1}{2}\sin(1)$
   (B) $\frac{1}{2\sqrt{2}}\sin(2)$
   (C) $\frac{1}{\sqrt{2}}\sin\left(\frac{1}{2}\right)$
   (D) $\frac{1}{4}\sin(4)$

64. If tangent at a point on the hyperbola $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$ cuts the axis at A and B and rectangle OAPB is completed, then locus of point P is given by (when O is the centre of the hyperbola)
   (A) $\frac{a^2}{x^2}-\frac{b^2}{y^2}=1$
   (B) $\frac{a^2}{y^2}+\frac{b^2}{x^2}=1$
   (C) $\frac{a^2}{y^2}-\frac{b^2}{x^2}=1$
   (D) $\frac{x^2}{a\sqrt{1-b^2}}+\frac{y^2}{b\sqrt{1-a^2}}=1$

65. Let $L_1 : \frac{x-1}{2} = \frac{y-2}{-1} = \frac{z-3}{3}$, $L_2 : \frac{x-1}{-1} = \frac{y-2}{3} = \frac{3(z-3)}{5}$ and $\frac{x-1}{-32} = \frac{y-2}{-19} - \frac{z-3}{15}$ be three lines. A plane intersecting these lines at A, B and C respectively such that PA = 2, PB = 3 and PC = 6 where P (1, 2, 3). If V is the volume of the tetrahedron PABC and d is the perpendicular distance of the plane from the point P then:
(A) V = 18 cubic units
(B) V = 8 cubic units
(C) $d = \frac{6}{\sqrt{14}}$ units
(D) $d = 7$ units

66. If $\frac{1}{\sqrt{x-1}} + \frac{1}{\sqrt{y-1}} + \frac{1}{\sqrt{z-1}} > 0$ and x, y, z are in G.P. then $\left(\log x^2\right)^{-1}, \left(\log xy\right)^{-1}$ and $\left(\log xz\right)^{-1}$ are in
(A) A.P.
(B) G.P.
(C) H.P.
(D) none of these

67. Let A and B are square matrices of same order satisfying AB = A and BA = B, then $\left(A^{2019} + B^{2019}\right)^{2020}$ is equal to :
(A) $A + B$
(B) $2020(A + B)$
(C) $2^{2019}(A + B)$
(D) $2^{2020}(A + B)$

68. Let $\vec{a}, \vec{b}, \vec{c}$ be three non – collinear vectors in a plane such that $|\vec{a}| = 1$, $|\vec{b}| = 3$ and $|\vec{c}| = \sqrt{10}$. If $\vec{a} \wedge \vec{c} = \alpha$ and $\vec{b} \wedge \vec{c} = \beta$ where $\alpha, \beta \in \left[\frac{\pi}{2}, \pi\right]$, then $\alpha + \beta$ equals:
(A) $\frac{7\pi}{6}$
(B) $\frac{4\pi}{3}$
(C) $\frac{3\pi}{2}$
(D) $\frac{7\pi}{4}$

69. Let 'P' be a point on the line $y = 3x$ and Q lies on $2y + x = 2$ and the mid – point of PQ is (2, 3). The distance PQ is
(A) 4
(B) 6
(C) 8
(D) 10

70. The coordinates of the middle point of the chord cut off by the line $2x-5y+18=0$ on the circle $x^2+y^2-6x+2y-54=0$ are
(A) (1, 4)
(B) (2, 4)
(C) (4, 1)
(D) (1, 1)

71. If $T_n=\frac{4n+\sqrt{4n^2-1}}{\sqrt{2n+1}+\sqrt{2n-1}}$, then $\sum_{n=1}^{40} T_n =$
(A) 360
(B) 720
(C) 365
(D) 364

72. If ${}^nC_0-{}^nC_1+{}^nC_2-{}^nC_3+.....+(-1)^r\,{}^nC_r=28,$ Then n is equal to
(A) 2
(B) 4
(C) 6
(D) 9

73. The number of 5 digit numbers of the form x y z y x in which x < y is
(A) 350
(B) 360
(C) 380
(D) 390

74. If $\sin A+\sin 2A=x$ and $\cos A+\cos 2A=y$, then $\left(x^2+y^2\right)\left(x^2+y^2-3\right)=$
(A) 2y
(B) y
(C) 3y
(D) none of these

75. The general solution of the equation $\tan x+\tan\left(x+\frac{\pi}{3}\right)+\tan\left(x+\frac{2\pi}{3}\right)=3$ is
(A) $\frac{n\pi}{4}+\frac{\pi}{12}, n\in I$
(B) $\frac{n\pi}{3}+\frac{\pi}{6}, n\in I$
(C) $\frac{n\pi}{3}+\frac{\pi}{12}, n\in I$
(D) none of these

76. If $2\tan^2 x - 5\sec x - 1 = 0$ has 7 different roots in $\left[0, \frac{n\pi}{2}\right]$, $n \in N$, then greatest value of n is
(A) 8
(B) 10
(C) 13
(D) 15

77. If $f(1) = -2\sqrt{2}$ and $g'(\sqrt{2}) = 4$, then the absolute value of the derivative of $f(\tan x)$ with respect to $g(\sec x)$ at $x = \frac{\pi}{4}$ is
(A) 4
(B) 3
(C) 2
(D) 1

78. Product of roots of equation $\begin{vmatrix} 1+2x & 1 & 1-x \\ 2-x & 2+x & 3+x \\ x & 1+x & 1-x^2 \end{vmatrix} = 0$ is p, then find the value of $|4p|$
(A) 2
(B) 4
(C) 6
(D) 8

79. Consider two planes $P_1 : 2x - 3y + z = -5$ and $P_2 : x - 4y - 14z = 5$. If a line L which is perpendicular to line of intersection of planes P_1 and P_2 whose one point $A(-7, y_1, z_1)$ lies on the plane P_1 and point $B(x_2, y_2, z_2)$ and $C(x_3, y_3, z_3)$ lie on plane P_2 has the equation $\frac{x+7}{a} = \frac{y-y_1}{-b} = \frac{z-z_1}{c}$, where $a, b, c \in N$, then:
(A) least value of $(a+b+c)$ is 5
(B) $y_1 + z_1$ is equal to –4
(C) $\frac{z_3 - z_2}{y_3 - y_2} = 3$
(D) $\frac{x_3 - x_2}{z_3 - z_2} = 2$

80. Let $\vec{a}, \vec{b}, \vec{c}$ be three vectors representing sides of a triangle such that $\vec{a} + \vec{b} + \vec{c} = 0$, $\vec{a} \wedge \vec{b} = 60^o$. If $|\vec{a}| = |\vec{b}|$ and $|\vec{c}| = 6$, then:
(A) area of the triangle is $9\sqrt{3}$
(B) area of the triangle is $3\sqrt{3}$
(C) length of the median bisecting the vector $\vec{c}$ is $4\sqrt{3}$
(D) length of the median bisecting the vector $\vec{c}$ is $3\sqrt{3}$

SECTION – B
(Numerical Answer Type)

This section contains **10** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

81. The number of statements which are false in the following:
$(s-1) p \wedge (\sim p)$ is a contradiction
$(s-2) (p \Rightarrow q) \Leftrightarrow (\sim q \Rightarrow \sim p)$ is a contradiction
$(s-3) \sim(\sim p) \Leftrightarrow p$ is a tautology
$(s-4) p \vee (\sim p) \Leftrightarrow$ is a tautology

82. Let equation of plane be $x+2y+z-3=0$. An insect starts flying from point P (1, 3, 2) in straight line. It touches the plane at point R (a, b, c) and then goes to point Q (3, 5, 2) in straight line. If distance travelled PQ + QR is minimum, then find the value of $(a+b+c)$.

83. The remainder when $\sum_{r=1}^{10} \left({}^{10}C_r\right)(r-5)^3$ is divided by 10, is

84. An urn contains 3 red, 4 green and certain number of white balls. Two balls are drawn simultaneously and found to be of different colour. If the chance that none of them was a white ball is $\frac{2}{9}$, then number of white balls is equal to

85. If $\vec{a}.\vec{b}$ are any two perpendicular vectors of equal magnitude and
$\left|3\vec{a}+4\vec{b}\right|+\left|4\vec{a}-3\vec{b}\right|=20$ then $\left|\vec{a}\right|$ equals:

86. Number of points at which
f(x) = $\sin\left([x]\frac{\pi}{2}\right)\operatorname{sgn}(x^2-3x+2)$
is discontinuous in $x=[0, 3]$, is (where [.] and sgn (.) denotes greatest integer and signum function respectively)

87. Let $A=\begin{bmatrix} a & x & p \\ y & q & b \\ r & c & z \end{bmatrix}$ and $B=\begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$ where a, b, c, x, y, z, p, q, r are natural numbers.
If tr.$\left(AB+AB^3+AB^5+....+AB^{19}\right)=210$, then find number of ordered triplets (p, q, r)
[Note : tr. (P) denotes the trace of matrix P.]

88. Sum of positive roots of the equation $\left(x^2-12x+35\right)\left(x^2+10x+24\right)=504$ is:

89. Let $A = \begin{bmatrix} x^3+1 & x+5 & 3x+2 \\ y+1 & -6x^2+2 & z-1 \\ 2 & 3 & 9x+6 \end{bmatrix}$ and $|A| = 3$

If $f(x) = \text{tr.}\left(B^{-1}\right)$ and $B = \text{adj}(A)$, then global maximum value of $f(x)$ in $x \in [0,6]$ is:

90. Let $D = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix}$ and $P = \begin{bmatrix} 7 & 0 & 2 \\ 0 & 1 & 0 \\ 2 & 0 & 5 \end{bmatrix}$. Consider $A = P^{-1}DP$. Find $\det\left(A^2 + A\right)$.